RFEM: Standard vs. Sink-Aware

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Methodology

Standard RFEM — four stages:

Stage A — Per-head rollout.
Each attention head h is rolled out separately across all L layers. Rows are re-normalised after adding the identity:

$$\hat{A}_h = \prod_{l=1}^{L} (A_h^{(l)} + I)$$

Stage B — K-σ filter.
Entries below the per-head threshold are zeroed out:

$$\tilde{A}_h(i,j) = \begin{cases} \hat{A}_h(i,j), & \hat{A}_h(i,j) \geq \mu_h + K\sigma_h \\ 0, & \text{otherwise} \end{cases}$$

where μ_h and σ_h are computed from the read-out row of $\hat{A}_h$.

Stage C — Weighted head aggregation.
Filtered per-head masks are combined using the per-head maximum as weight:

$$A_{\text{rfem}} = \sum_{h=1}^{H} w_h \, \tilde{A}_h, \qquad w_h = \max(\hat{A}_h)$$

Stage D — Read-out.
Importance scores are extracted from the read-out row of A_{rfem}: BERT $\to$ `[CLS]` row $\to$ token bars; CLIP-Text $\to$ `[EOS]` row $\to$ token bars; CLIP-Vision $\to$ `[CLS]` row $\to$ 7×7 patch heatmap.

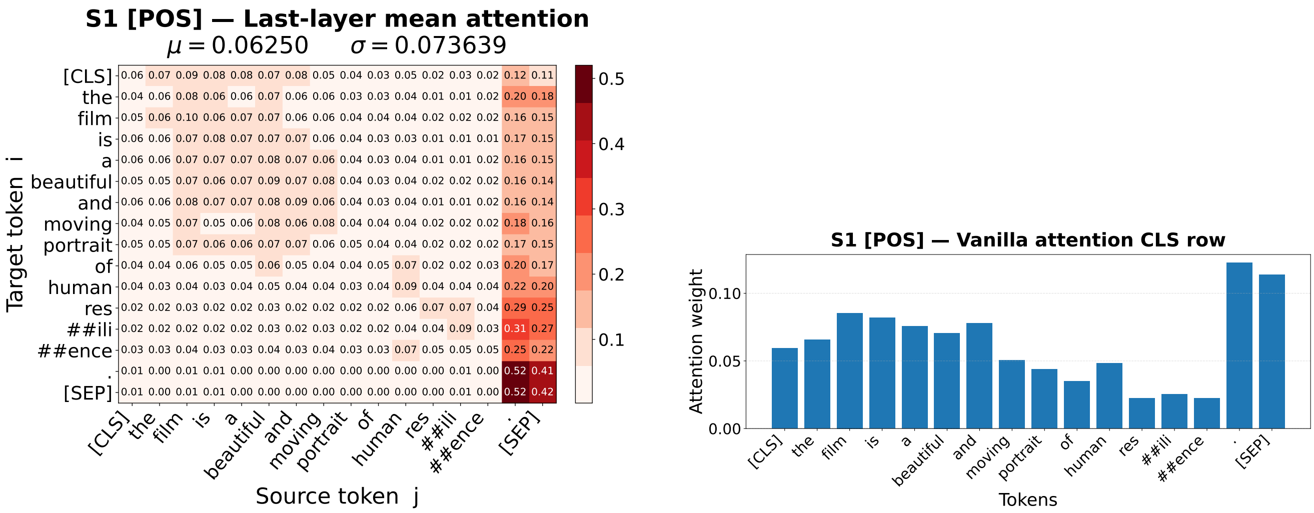
Sink-aware variant:

Problem.
Certain structural tokens act as *attention sinks* — they absorb disproportionate probability mass: `[CLS]`/`[SEP]` in BERT, BOS/EOS in CLIP-Text, and the CLS-self position in CLIP-Vision. Sink entries inflate σ_h, pushing the threshold above all content tokens and zeroing them out.

Fix.
μ_h and σ_h are computed from the read-out row *excluding the sink token columns*. The threshold is then calibrated to the content-token distribution only, so content tokens are no longer suppressed.

BERT — Sentence S1

Baseline: Vanilla Attention



(a) Last-layer mean attention matrix (all 12 heads averaged). `[CLS]` and `[SEP]` columns attract the dominant mass.

(b) Vanilla CLS read-out scores. Sink tokens `[CLS]` and `[SEP]` dominate; all content words are suppressed.

Figure 1. Vanilla attention (BERT, S1). Raw last-layer attention already shows severe sink behaviour before any rollout or filtering.

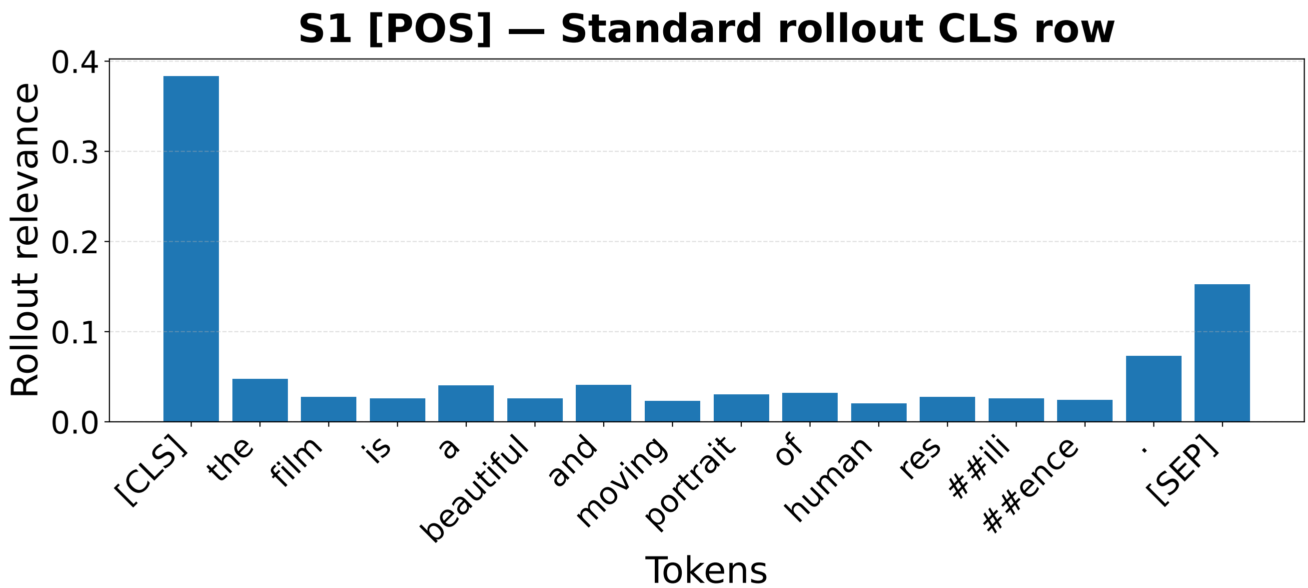
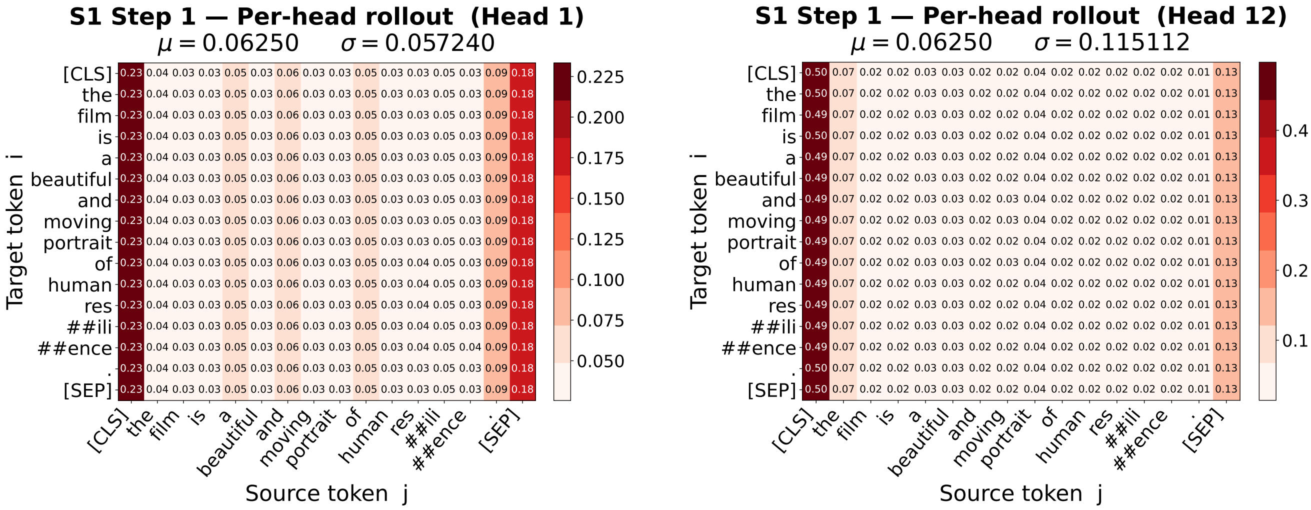


Figure 2. Standard attention rollout — CLS row scores (BERT, S1). Multiplicative rollout across all 12 layers amplifies the sink effect: `[CLS]` and `[SEP]` absorb virtually all weight, leaving content tokens near zero.

RFEM — Stage A: Per-Head Rollout

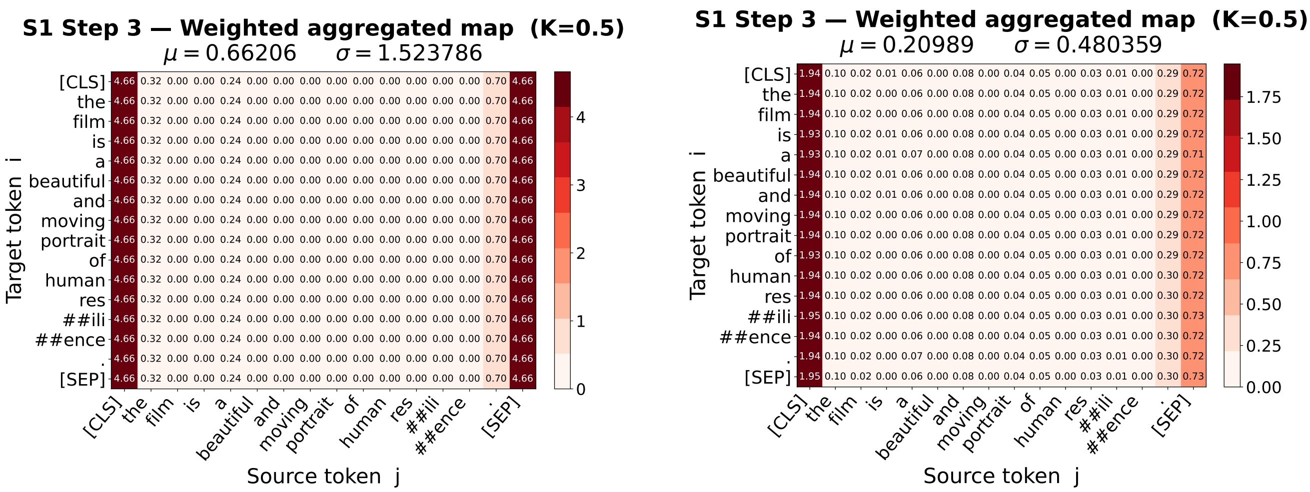


(a) Head 1 rolled-out matrix $\hat{A}_{h=1}$. The `[CLS]` column (leftmost) is the dominant sink.

(b) Head 12 rolled-out matrix $\hat{A}_{h=12}$. Both `[CLS]` and `[SEP]` columns concentrate the weight.

Figure 3. Per-head rollout matrices (BERT, S1). Every head independently accumulates sink mass through all 12 layers. The K-σ filter must overcome this inflated baseline to surface content tokens.

RFEM — Stage C: Weighted Head Aggregation

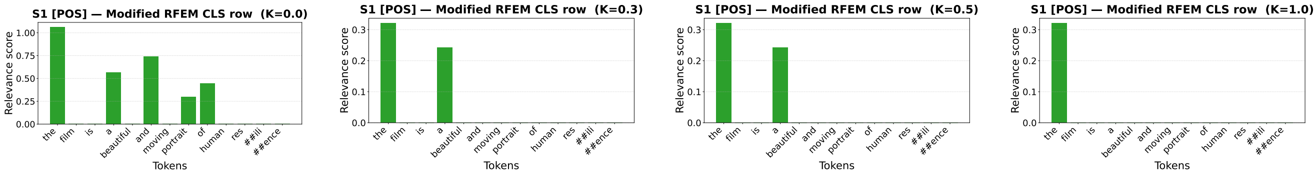


(a) Standard RFEM A_{rfem} (K=0.5). Sink columns still carry dominant weight after filtering.

(b) SA-RFEM A_{rfem} (K=0.5). Sink columns suppressed; content token columns are recovered.

Figure 4. Weighted aggregation matrix — Standard vs. Sink-Aware RFEM at K=0.5 (BERT, S1). In the sink-aware variant, μ_h and σ_h are estimated from content-only CLS row entries, making the threshold meaningful for content tokens.

RFEM — Stage D: Token Importance Scores (Standard RFEM)



(a) Standard RFEM K=0.0 **(b)** Standard RFEM K=0.3 **(c)** Standard RFEM K=0.5 **(d)** Standard RFEM K=1.0

Figure 5. Standard RFEM — CLS token scores for all K values (BERT, S1). `[CLS]`/`[SEP]` sink tokens retain high scores at every threshold level. Increasing K aggressively zeros out content tokens rather than filtering the sinks.

RFEM — Stage D: Token Importance Scores (Sink-Aware RFEM)

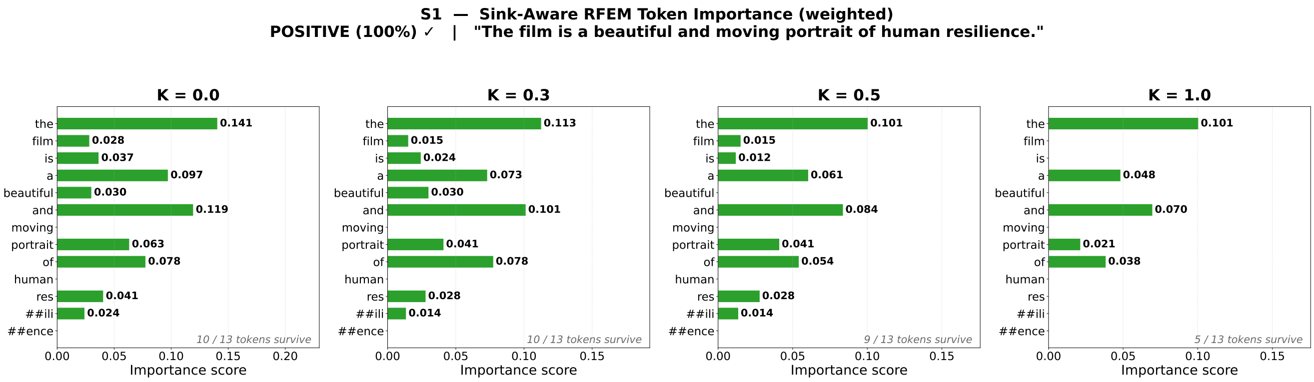
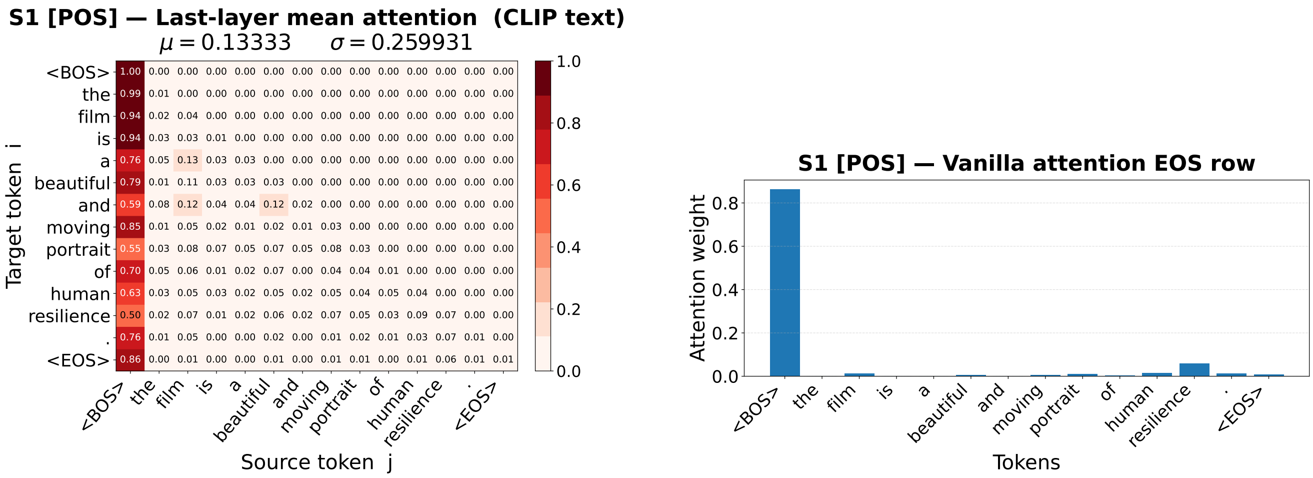


Figure 6. Sink-Aware RFEM — CLS token importance across all K values (BERT, S1). With sink columns excluded from threshold estimation, content words such as *beautiful*, *resilience*, and *portrait* surface with meaningful raw scores. Higher K progressively removes low-confidence tokens while the most salient content words remain stable.

CLIP-Text Encoder — Sentence S1

Baseline: Vanilla Attention



(a) Last-layer mean attention matrix. The BOS column (far left) and EOS diagonal entry absorb the dominant mass in every row.

(b) Vanilla EOS read-out scores. BOS and EOS tokens monopolise the read-out row; content tokens are negligible.

Figure 7. Vanilla attention (CLIP-Text, S1). Causal masking concentrates the EOS row on BOS — the structural sink — before any RFEM processing.

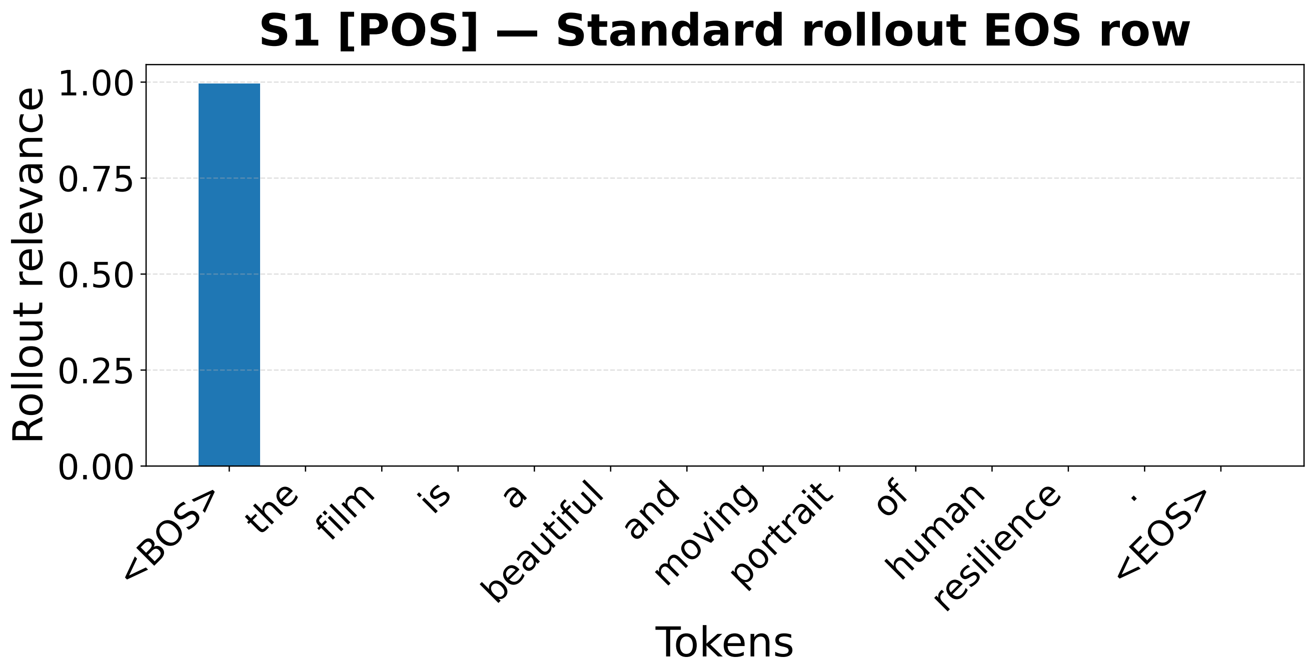
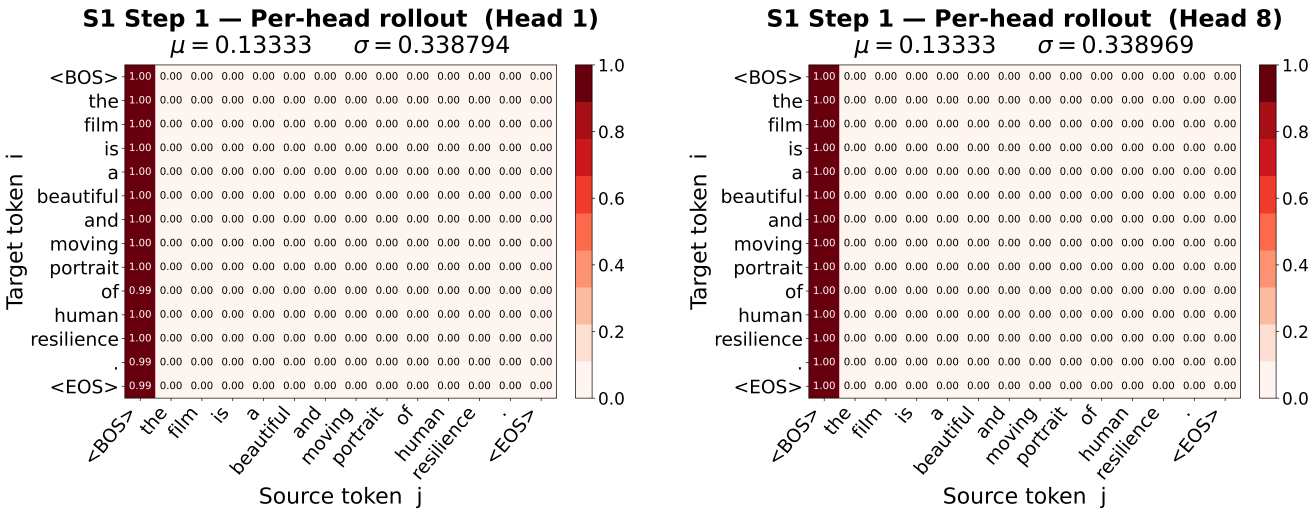


Figure 8. Standard attention rollout — EOS row scores (CLIP-Text, S1). Layer-by-layer rollout amplifies BOS sink dominance. Content tokens are entirely suppressed, making standard rollout uninformative for text interpretation.

RFEM — Stage A: Per-Head Rollout

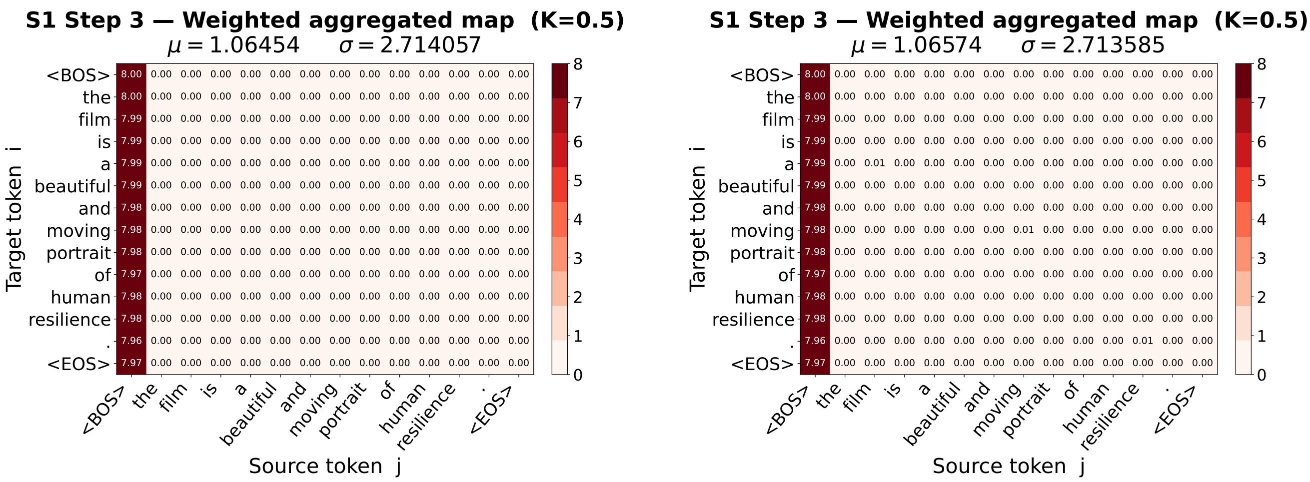


(a) Head 1 rolled-out matrix $\hat{A}_{h=1}$. BOS column (col 0) concentrates the largest weight mass.

(b) Head 8 rolled-out matrix $\hat{A}_{h=8}$. Both BOS and EOS positions act as sinks in this head.

Figure 9. Per-head rollout matrices (CLIP-Text, S1). All 8 heads exhibit BOS-dominated columns after full rollout. The inflated σ_h in standard RFEM prevents content tokens from surviving the K-σ filter.

RFEM — Stage C: Weighted Head Aggregation

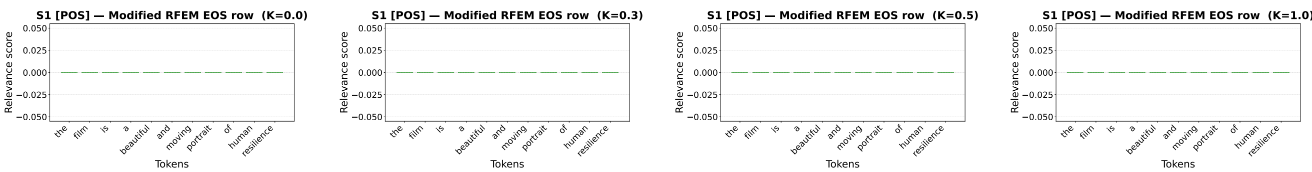


(a) Standard RFEM A_{rfem} (K=0.5). BOS column still dominates; content columns are near-zero.

(b) SA-RFEM A_{rfem} (K=0.5). BOS/EOS excluded from threshold — content columns become visible.

Figure 10. Weighted aggregation matrix — Standard vs. Sink-Aware RFEM at K=0.5 (CLIP-Text, S1). The sink-aware threshold is calibrated against content-token values in the EOS row only, enabling content signal to propagate through the filter.

RFEM — Stage D: Token Importance Scores (Standard RFEM)



(a) Standard RFEM K=0.0

(b) Standard RFEM K=0.3

(c) Standard RFEM K=0.5

(d) Standard RFEM K=1.0

Figure 11. Standard RFEM — EOS token scores for all K values (CLIP-Text, S1). All content tokens score zero at every threshold because the BOS sink inflates σ_h beyond the content-token scale. Standard RFEM completely fails for CLIP-Text.

RFEM — Stage D: Token Importance Scores (Sink-Aware RFEM)

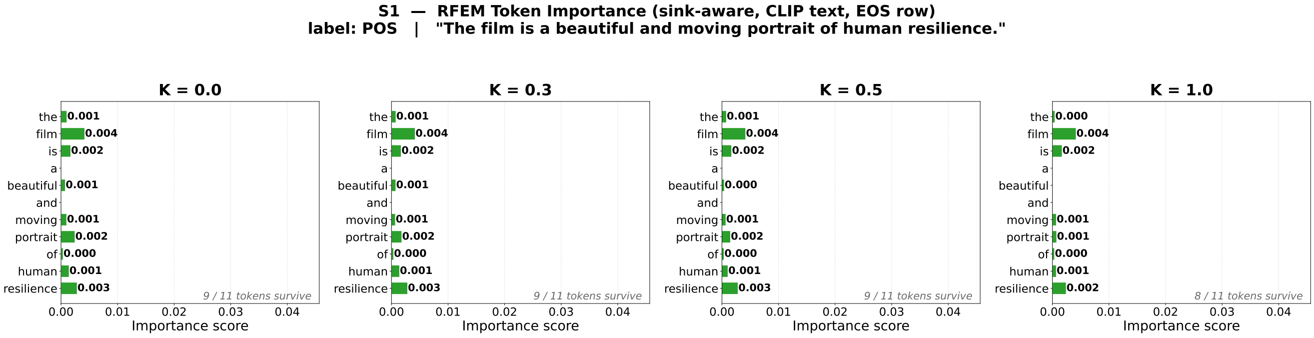


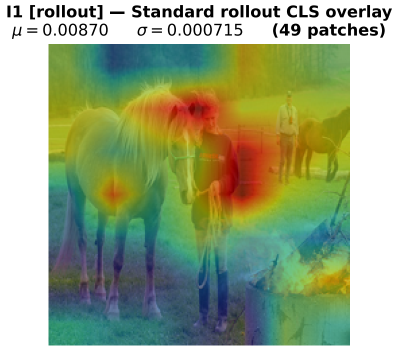
Figure 12. Sink-Aware RFEM — EOS token importance across all K values (CLIP-Text, S1). Meaningful content words such as *film*, *portrait*, and *resilience* emerge with raw unnormalised scores after sink exclusion. Higher K progressively focuses on the most distinctive tokens.

CLIP-Vision Encoder — Image I1

Baseline: Original Image, Vanilla Attention & Rollout

(a) Original input image. Clear foreground subject with structured background.

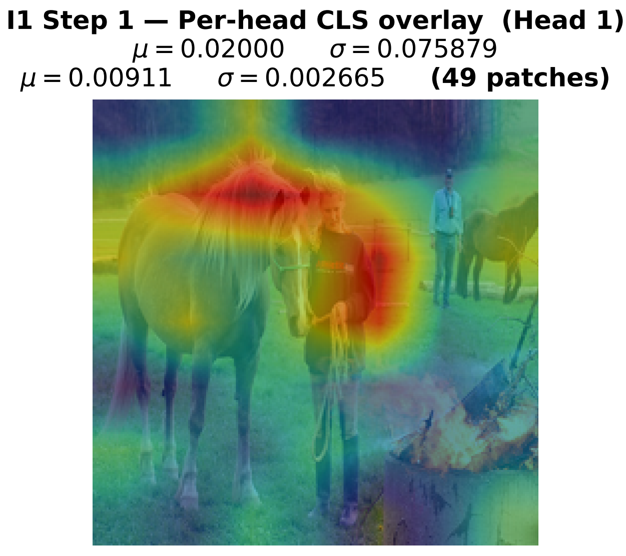
(b) Vanilla CLS attention overlay (last layer, all heads averaged). Near-uniform heatmap — the CLS-self sink at position (0,0) spreads weight indiscriminately across all patches.



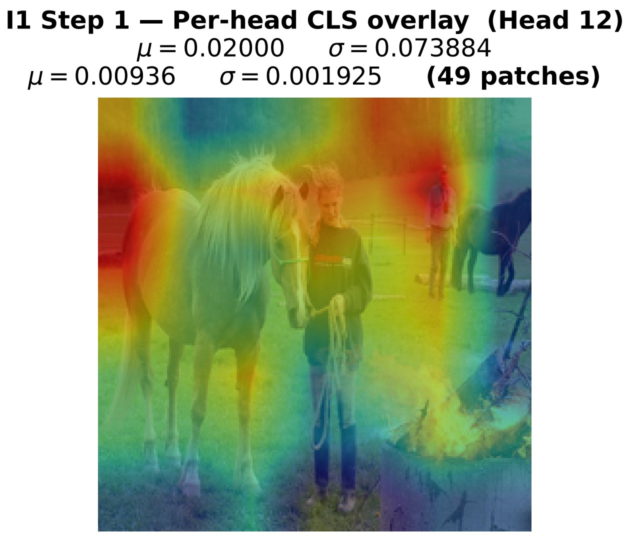
(c) Standard rollout overlay. Sink dominance compounds layer by layer, yielding a diffuse, spatially non-discriminative heatmap.

Figure 13. Baseline visualisations (CLIP-Vision, I1). Before RFEM, neither vanilla attention nor standard rollout provides meaningful spatial localisation — the CLS-self sink overwhelms content patches at every layer.

RFEM — Stage A: Per-Head Rollout



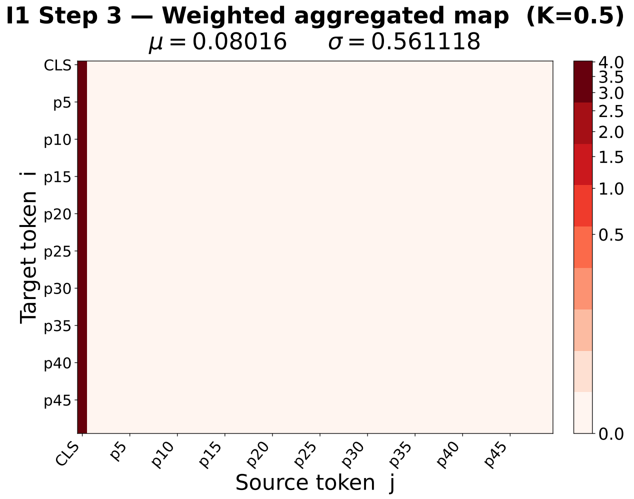
(a) Head 1 rolled-out overlay $\hat{A}_{h=1}$ (CLS row). Spatial structure begins to emerge; attention concentrates on parts of the scene.



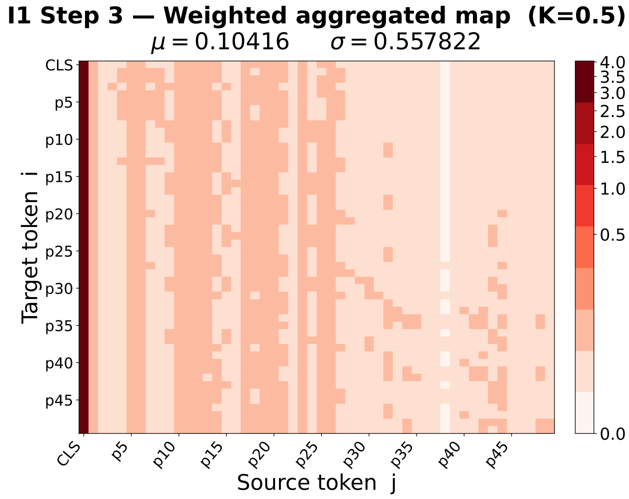
(b) Head 12 rolled-out overlay $\hat{A}_{h=12}$ (CLS row). This head specialises in a different spatial region — complementary to Head 1.

Figure 14. Per-head rollout overlays — Stage A (CLIP-Vision, I1). Individual heads encode diverse spatial attention patterns after full layer-by-layer rollout. These per-head maps feed into the K-σ filter (Stage B) and weighted aggregation (Stage C). Both Standard and SA-RFEM share the same per-head rollout at this stage.

RFEM — Stage C: Weighted Head Aggregation Matrix



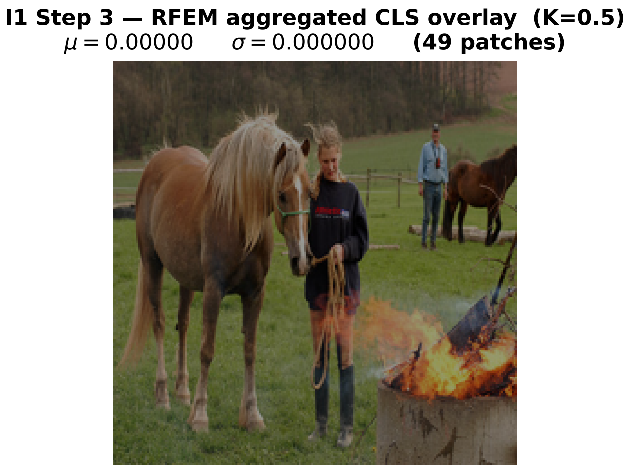
(a) Standard RFEM aggregation matrix A_{rfem} (K=0.5). The CLS row (top row) shows diffuse weight — sink inflates the threshold and most content entries are zeroed.



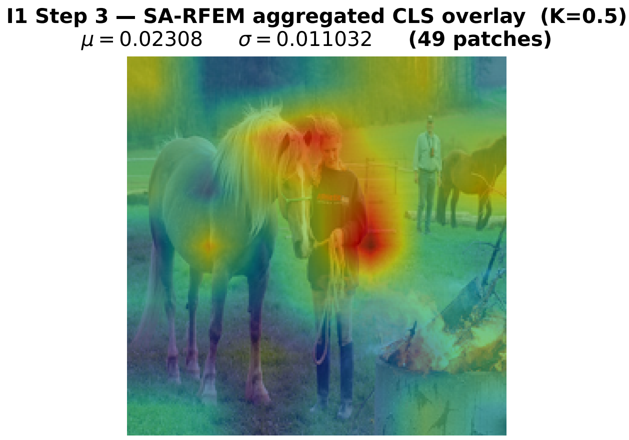
(b) SA-RFEM aggregation matrix A_{rfem} (K=0.5). With CLS-self excluded from threshold calibration, content patch entries in the CLS row survive and become concentrated.

Figure 15. Weighted aggregation matrix — Standard vs. Sink-Aware RFEM at K=0.5 (CLIP-Vision, I1). The matrix shows the full 50×50 aggregated attention map. The CLS row (used for read-out) is clearly more structured in SA-RFEM.

RFEM — Stage C: Weighted Head Aggregation Overlay



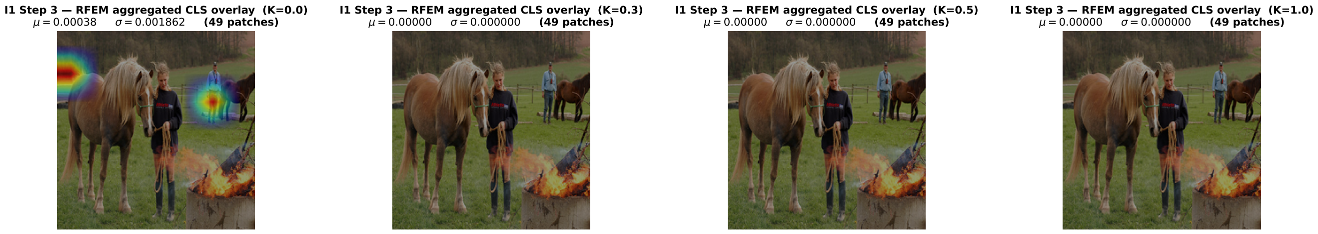
(a) Standard RFEM overlay (K=0.5). Heatmap remains diffuse over the image — the sink-inflated threshold suppresses true content patches.



(b) SA-RFEM overlay (K=0.5). CLS-self position excluded from threshold estimation — content patches are strongly and tightly highlighted on the foreground subject.

Figure 16. Weighted aggregation overlay — Standard vs. Sink-Aware RFEM at K=0.5 (CLIP-Vision, I1). The sink-aware approach produces a concentrated, semantically meaningful jet heatmap localised on the foreground subject, in stark contrast to the diffuse standard result.

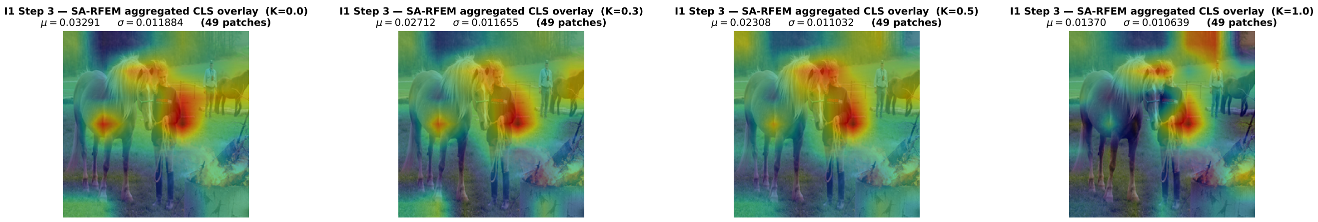
RFEM — Stage D: K-σ Sweep Overlays (Standard RFEM)



(a) Standard RFEM K=0.0 **(b)** Standard RFEM K=0.3 **(c)** Standard RFEM K=0.5 **(d)** Standard RFEM K=1.0

Figure 17. Standard RFEM — jet heatmap overlays across all K values (CLIP-Vision, I1). The heatmap remains diffuse at every sparsity level because the CLS-self sink inflates σ_h, causing the K-σ filter to suppress content patches regardless of the chosen K.

RFEM — Stage D: K-σ Sweep Overlays (Sink-Aware RFEM)



(a) SA-RFEM K=0.0 **(b)** SA-RFEM K=0.3 **(c)** SA-RFEM K=0.5 **(d)** SA-RFEM K=1.0

Figure 18. Sink-Aware RFEM — jet heatmap overlays across all K values (CLIP-Vision, I1). At K=0.0 all content patches survive; increasing K progressively tightens the jet heatmap onto the most salient foreground region, demonstrating clean controlled sparsity without sink distortion.